

A Research Gap: Potential Multi-Pathogen Transmission via Corrugated Cardboard

Hanyu Wang

Background

Corrugated cardboard, as the most commonly used material for package packaging, has been found to potentially increase the risk of infectious disease transmission by carrying viruses on its surface (Zambrana & Boehm, 2023) and harboring parasites or parasite eggs internally (Dolhun & Antes, 2016). Although many studies have examined pathogen transmission via fomites, these investigations typically focus on a single type of pathogen, and there is currently a lack of systematic research addressing the potential transmission risks posed by the complex context of corrugated cardboard. Meanwhile, with the rapid growth of international trade and cross-regional shopping, the widespread use of packages may further amplify the risk of disease transmission. Therefore, studying the potential of corrugated cardboard to carry multiple pathogens is important for understanding its role in disease transmission and assessing associated public health risks.

Objective

Investigating which diseases can potentially be transmitted via corrugated cardboard packages can provide critical guidance for logistics and warehousing practices, inform risk-reduction strategies in the handling and transportation of packages, and promote safer consumer behaviors, such as proper hand hygiene and package disinfection. Such research can also support evidence-based policies for the management of cross-regional and international parcel distribution, ultimately contributing to the prevention and control of infectious disease spread.

Reference

- Dolhun, E. P., & Antes, A. W. (2016). A case of cardboard boxes likely facilitating the biting of a patient by trypanosoma cruzi-infected triatomine bugs. *The American Society of Tropical Medicine and Hygiene*, 95(5), 1115–1117. <https://doi.org/10.4269/ajtmh.16-0455>
- Zambrana, W., & Boehm, A. B. (2023). Occurrence of human viruses on fomites in the environment: A systematic review and meta-analysis. *ACS Environmental Au*, 3(5), 277–294. <https://doi.org/10.1021/acsenvironau.3c00025>